

Dr. Georgii Oblapenko

Born on 10 Feb. 1992

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Languages: English (fluent), German (fluent), Russian (native speaker)



Researcher with 10+ years of experience in the field of numerical model development for simulation of non-equilibrium flows.

Research interests

- Mathematical modelling of multi-species non-equilibrium flows
- Particle-based methods for rarefied flows
- Discrete velocity methods
- Entropy-stable high-order methods
- Optimization in context of kinetic theory
- Research software development

Employment history

- Nov. 2023 – **Present** **Member of Scientific Staff** • RWTH Aachen, Chair of Applied and Computational Mathematics.
Development of 1) kinetic methods for modelling of non-equilibrium flows in continuum and rarefied regimes 2) entropy-stable DG schemes for multi-component real gas flows with thermo-chemical non-equilibrium. Development and teaching of new courses for Master students, teaching of global exercises for Bachelor students.
- Jan. 2023 – Sep. 2023 **Scientific Consultant** • UT Austin, Oden Institute for Computational Engineering and Sciences / Dept. of Aerospace Engineering and Engineering Mechanics.
Development of new PIC-DSMC methods. Co-supervision of Master student developing PIC-DSMC methods for high-voltage gap closure modeling.
- Nov. 2021 – Oct. 2023 **Postdoctoral Researcher** • German Aerospace Center (DLR), Institute of Aerodynamics and Flow Technology, Spacecraft Department.
Uncertainty quantification of hypersonic expanding continuum flows. Development of 1) an efficient photon Monte Carlo radiation transport code 2) novel kinetic methods for simulations of plasma and gases with internal degrees of freedom 3) low-rank methods for the Boltzmann equation. Supported by a personal grant of the Alexander von Humboldt Foundation.
- July 2018 – June 2021 **Postdoctoral Researcher** • UT Austin, Oden Institute for Computational Engineering and Sciences / Dept. of Aerospace Engineering and Engineering Mechanics.
Development of 1) hybrid numerical methods for kinetic problems 2) new algorithms for rarefied plasma simulations 3) novel kinetic models for gases with internal degrees of freedom and chemically reacting gas flows.

Employment history (continued)

Sept. 2013 – Dec. 2017  **Research engineer** • Saint-Petersburg State University, Department of Hydroaeromechanics.
Lead developer of a C++ library for kinetic theory computations and coupling with CFD codes. Development of 1) theoretical models of reaction rates in viscous gas flows and their numerical modeling 2) simplified models for vibrational relaxation rates.

Education

- 2015 – 2017  **Ph.D., Saint-Petersburg State University** in Physical and Mathematical Sciences.
Area of specialization: Mechanics of Fluids, Gases and Plasma.
Thesis title: *Physico-chemical relaxation rates in viscous non-equilibrium gas flows*.
Supervisor: Prof. Kustova E.V.
- 2013 – 2015  **M.Sc. (summa cum laude), Saint-Petersburg State University** in Mechanics and Mathematical Modeling.
Area of specialization: Molecular-kinetic theory of fluids and gases.
Thesis title: *Vibrational relaxation rates in viscous rarefied gas flows*.
Supervisor: Prof. Kustova E.V.
- 2009 – 2013  **B.Sc., Saint-Petersburg State University** in Mathematics and Mechanics.
Supervisor: Prof. Kustova E.V.

Grants and awards

Personal funding and travel grants

- 2026 – 2029  **DFG individual grant**. “Simulation of rare and transient events in rarefied gas dynamics”.
Funding of a doctoral position for 3 years (ca. 330000 EUR)
- 2021 – 2023  **Alexander von Humboldt Foundation** Research Stipend for Postdoctoral Researchers (60000 EUR).
- 2017 – 2018  **German Academic Exchange Service (DAAD)** Research Stipend (3300 EUR).
- 2016  **Santander Bank** Research Stipend (1400 EUR).
- 2015 – 2016  **Russian Foundation for Basic Research** Research Stipend (4500 EUR).
- 2013, 2014, 2017  **Saint-Petersburg State University** Travel funding awards.

Awards

- 2023  **Marie Skłodowska-Curie Postdoctoral Action Seal of Excellence**
- 2017  **Stipend** of the Russian President for students and PhD students in prioritized study programmes (3800 EUR).
- 2016  **Stipend** of the Russian Government for students and PhD students in prioritized study programmes (1900 EUR).
-  **Stipend** of the Russian Government for students and PhD students (500 EUR).
- 2013  **Winner** of the Saint-Petersburg Government grant competition for undergraduate and graduate students (500 EUR).

Publications

The full list of publications consists of **31** (including accepted papers) peer-reviewed publications in SCOPUS/Web of Science-indexed journals and conference proceedings, as well as **21** publications without

peer review (conference proceedings, preprints).

Publication statistics (as of July 2026):

	SCOPUS	Google Scholar
Number of publications	35	66
h-index	11	13
Citations	398	658

Conference and seminar presentations

- Conferences  Presented at **26** international and **11** national (German, Russian) conferences.
- Plenary talks  **2** invited plenary lectures at international conferences and workshops.
- Invited seminar talks  **27** invited talks at universities and research institutes in the USA, Germany, Belgium, France, and the Netherlands.
- Public outreach and pop-science talks  Lightning Math Night (RWTH Aachen, 2026)  Exhibition “Falling Rhythm” (Barcelona, Spain & Online, 2025)

Teaching and student supervision

Teaching: lectures

- Summer semester 2026  Development and teaching of new Master course on “Modern Simulation Software Development”, RWTH Aachen (together with Dr. Lambert Theisen). Course offered to Master and Bachelor students of the Computational Engineering Science and Simulation Sciences programs.
- Winter semester 2025/2026  Invited lecture on kinetic theory in the framework of the Master course on “Aerospace Fundamentals”, Justus Liebig University Gießen.
- Summer semester 2025  Development and teaching of new Master course on “Particle-based Simulation Methods”, RWTH Aachen. Course offered to Master and Bachelor students of the Mathematics, Computational Engineering Science, and Simulation Sciences programs.
- Winter semester 2024/2025  Invited lecture on kinetic theory in the framework of the Master course on “Aerospace Fundamentals”, Justus Liebig University Gießen.
- Winter semester 2023/2024  Invited lecture on kinetic theory in the framework of the Master course on “Aerospace Fundamentals”, Justus Liebig University Gießen.

Teaching: exercises

- Winter semester 2025/2026  Global Exercise for “Foundations of Mathematics I”, Bachelor Study Program “Computational Engineering Science”, RWTH Aachen.
- Summer semester 2024  Global Exercise for “Foundations of Mathematics II”, Bachelor Study Program “Computational Engineering Science”, RWTH Aachen.
- Winter semester 2023/2024  Global Exercise for “Foundations of Mathematics I”, Bachelor Study Program “Computational Engineering Science”, RWTH Aachen.

Additional teaching qualifications

- 2026  Enrolled in program for “Extensions Module” certificate of the program “Professionelle Lehrkompetenz für die Hochschule”, Hochschuldidaktik NRW, expected ending date: August 2026

Teaching and student supervision (continued)

- 2025  “Basics Module” certificate of the program “Professionelle Lehrkompetenz für die Hochschule”, Hochschuldidaktik NRW
- 2024–2026  15 Completed courses at the Center for Teaching and Learning Services, RWTH Aachen.
- 2020  Completed Title IX Training on prevention and reporting of sexual discrimination and harassment, UT Austin.

Supervision, thesis reviews

- 2025 – **Present**  Regular supervision of Bachelor theses at RWTH Aachen.
Supervision of seminar and group research projects at RWTH Aachen.
Referee of Bachelor theses, RWTH Aachen.
- 2024 – **Present**  Regular supervision of student research assistants at RWTH Aachen.
Co-supervision of PhD Student Eda Yilmaz.
- 2022 – 2023  Co-supervision of a Master thesis at UT Austin.
- 2020 – 2021  Referee of Bachelor-, Diploma- and Master theses at Saint-Petersburg State University and Peter the Great St. Petersburg Polytechnic University.

Professional membership and other activities

- 2025 – **Present**  **RWTH hiring committee** • Member of the hiring committee for the W2 professorship “Industrial mathematics”, RWTH Aachen.
- March 2022 – March 2024  **COST Network** • Member of the networking group “Mathematical models for interacting dynamics on networks” (MAT-DYN-NET), supported by the COST (European Cooperation in Science and Technology) program.
- July 2021 – **Present**  **RGD NextGen Group** • Member of the RGD NextGen Group, aimed at increasing outreach of the Rarefied Gas Dynamics community.
- Spring 2021 – Jan. 2024  **American Institute of Aeronautics and Astronautics (AIAA) Thermophysics Technical Committee** • Member of Education, Emerging Technologies, Conference subcommittees.
- Dec. 2019 – Jan. 2024  **American Institute of Aeronautics and Astronautics (AIAA)** • Senior AIAA Member .

Editing and review activities

- Editor  Proceedings of the 33rd International Symposium on Rarefied Gas Dynamics, Springer, 2026 (co-editor), [ISBN 978-3-032-00093-1](#).
- Funding Proposal reviewer  DFG (since 2026).
- Journal reviewer  Computers and Fluids (Elsevier) · Acta Astronautica (Elsevier) · Journal of Computational Physics (Elsevier) · Computer Methods in Applied Mechanics and Engineering (Elsevier) · Applied Mathematical Modelling (Elsevier) · Physics of Fluids (AIP) · Physics of Plasmas (AIP) · Journal of Thermophysics and Heat Transfer (AIAA) · Journal on Scientific Computing (SIAM) · Microfluidics and Nanofluidics (Springer).
- Proceedings and conference reviewer  International Symposium on Rarefied Gas Dynamics · STAB/DGLR Symposium · AIAA Scitech · AIAA Aviation

Professional membership and other activities (continued)

Event organisation

- RGD33 Symposium  Member of the Local Advisory Committee of the “33rd International Symposium on Rarefied Gas Dynamics”, 15-19 July 2024, Göttingen, Germany.
- VIII Polyakhov Readings  Member of the Local Organizing Committee of the international conference on mechanics “VIII Polyakhov Readings”, 30 January-2 February 2018, Saint-Petersburg, Russia.

Research software

Developed open-source simulation software includes

- [A variable-weight Direct Simulation Monte Carlo code](#)
- [Sparse entropic quadrature method for gas-kinetic problems](#)
- [Extensions of Trixi.jl to aerodynamic applications](#)
- [Reproducible examples for the course *Modern Simulation Software Development*, RWTH Aachen](#)
- [Extensions of Trixi.jl to entropy-conservative simulations of reacting multi-species gas flows](#)
- [A library for calculation of thermodynamic and transport properties of non-equilibrium gas flows](#)
- [A tool for calculation of vibrational relaxation times based on a kinetic theory definition](#)